Melik Jabnouni

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Objective

Recently graduated Electrical Engineer, specialized in Embedded Systems, Industrial IoT, and Applied AI. Motivated to learn, adapt, and contribute to innovative engineering projects by developing end-to-end intelligent solutions combining hardware, software, and AI.

Education

National Engineering School of Monastir (ENIM) *2022 – 2026*
Engineering Degree in Electrical Engineering, Electrical Systems & Renewable Energies track. *Monastir - Tunisia*

Faculty of Sciences of Monastir (FSM) *2019 – 2022*
Preparatory Cycle for Engineering Studies, Physics-Chemistry (PC) track. *Monastir - Tunisia*

Professional Experience

Final-Year Internship (PFE), Photocarb *Feb. 2026 – June 2026*
Industrial IoT, Applied AI, Full-Stack Development *Tunisia*

- Designed an end-to-end intelligent industrial emissions monitoring platform, combined with an ESP32-based IoT sensor network.
- Implemented an LSTM forecasting model, achieving per-variable performance score gains of up to +24% over a persistence baseline, coupled with an Isolation Forest module for automated anomaly detection.
- Developed a multi-role MERN supervision dashboard with automated KPI reporting and multi-channel alerts via dedicated AlertService and KPIService modules.
- Positioned the solution against established industrial monitoring systems, at roughly 80-90% lower cost.
- Set up an MQTT/Mosquitto communication architecture between sensor stations and the back-end for reliable real-time data ingestion.

Technical Environment: ESP32, Python, TensorFlow/Keras, MERN stack, FastAPI, MQTT, Mosquitto, Docker, Git.

Embedded IoT Intern, Methane Energy Production Solutions *July 2025 – Aug. 2025*
Embedded Systems, IoT, Green Energy *Tunisia*

- Developed a real-time biogas monitoring system for green energy applications using an ESP32 microcontroller, ensuring continuous monitoring of the anaerobic digestion process.
- Implemented MQTT-based communication via AWS IoT Core, ensuring reliable and scalable transmission of sensor data to the cloud.
- Contributed to sensor integration and calibration for gas concentration and process parameter monitoring in an industrial biogas environment.

Technical Environment: ESP32, Embedded C, MQTT, AWS IoT.

Electrical Systems Intern, TRS BK SARL *July 2024 – Aug. 2024*
Electrical Networks, Compliance Auditing *Tunisia*

- Installed and audited residential and industrial electrical networks in compliance with the NFC 15-100 standard.
- Diagnosed non-conformities and proposed corrective actions to ensure regulatory compliance and installation safety.

Technical Environment: Electrical Installation, NFC 15-100, Circuit Diagnostics.

Personal and Academic Projects

Automatic Tunisian License Plate Recognition *2025*

- Developed a real-time detection and character recognition (OCR) pipeline for Tunisian Arabic/Latin license plates, using YOLOv8 for detection and OCR for character recognition.
- Deployed the pipeline on a Raspberry Pi 4 with an ESP32 camera module, including plate category classification for downstream analytics.

Technical Environment: YOLOv8, OpenCV, Python, Raspberry Pi 4, ESP32, OCR.

PCB Defect Detection & Component Identification *2025*

- Developed a computer vision pipeline for automated printed circuit board inspection, training custom detection models on Roboflow for defect localization.
- Integrated SAM (Segment Anything Model) for component segmentation and identification, improving inspection granularity.

Technical Environment: Roboflow, SAM, Python, OpenCV, Deep Learning.

Intelligent Football Match Analysis System *2025*

- Designed a computer vision pipeline combining YOLOv5 and CNNs to detect players and the ball in match footage.
- Applied KMeans clustering for team assignment and generated structured performance statistics with visual analytics.

Technical Environment: YOLOv5, CNNs, KMeans, OpenCV, Python.

Square Wave Signal Generator – PCB & Embedded Firmware *2024*

- Designed a multi-mode signal generator built around the ATmega328P, producing three square-wave signal outputs simultaneously (NCO, PWM, Servo).
- Developed the embedded firmware handling menu navigation, parameter editing, and a locking mechanism, with user feedback via LCD screen, LEDs, and rotary encoder.
- Designed the complete schematic and PCB layout in Altium Designer, then manufactured the board in-house via UV exposure, chemical etching, drilling, and cold tinning.

Technical Environment: ATmega328P, Embedded C, Altium Designer, PCB Fabrication.

Hand Gesture LED Control – ESP32 & Python *2024*

- Implemented a real-time gesture recognition system with MediaPipe, controlling wireless actuators via ESP32.
- Designed an IoT-based human-machine interface for intuitive, contactless device control.

Technical Environment: ESP32, Python, MediaPipe, WiFi.

Skills

- **Programming:** Python, C/C++, Embedded C, JavaScript
- **Machine Learning & CV:** TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLOv5/v8, LSTM, CNNs, Roboflow, SAM
- **Embedded & IoT:** ESP32, STM32, PIC, Arduino, Raspberry Pi, MQTT, AWS IoT, Proteus, Wokwi
- **Web & Backend:** React.js, Node.js, Express.js, MongoDB, REST APIs
- **Tools:** Git, VS Code, STM32CubeMX, Qt Creator, Docker
- **Databases:** MongoDB
- **Languages**: Arabic (Native) French (Fluent) English (Fluent)